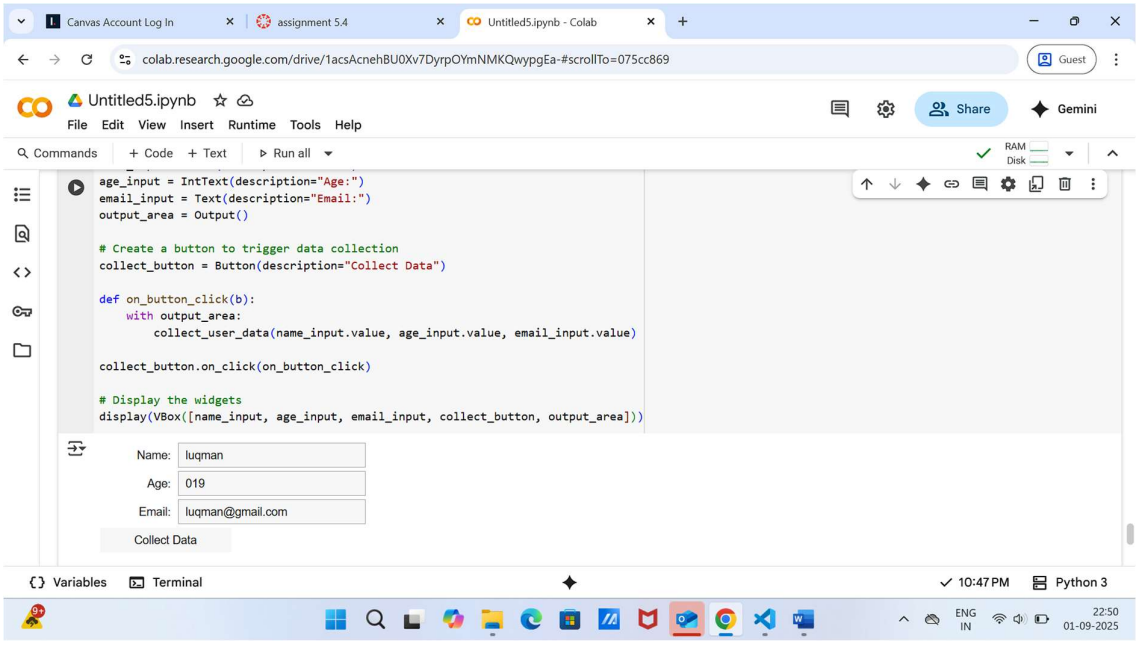
Canvas Account Log In
assignment 5.4
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age_input = IntText(description="Age:")
email_input = Text(description="Email:")
output_area = Output()
# Create a button to trigger data collection
collect_button = Button(description="Collect Data")
def on_button_click(b):
with output_area:
collect_user_data(name_input.value, age_input.value, email_input.value)
collect_button.on_click(on_button_click)
# Display the widgets
display(VBox([name_input, age_input, email_input, collect_button, output_area]))
Name: luqman
Age: 019
Email: luqman@gmail.com
Collect Data
Variables
Terminal
10:47 PM
Python 3
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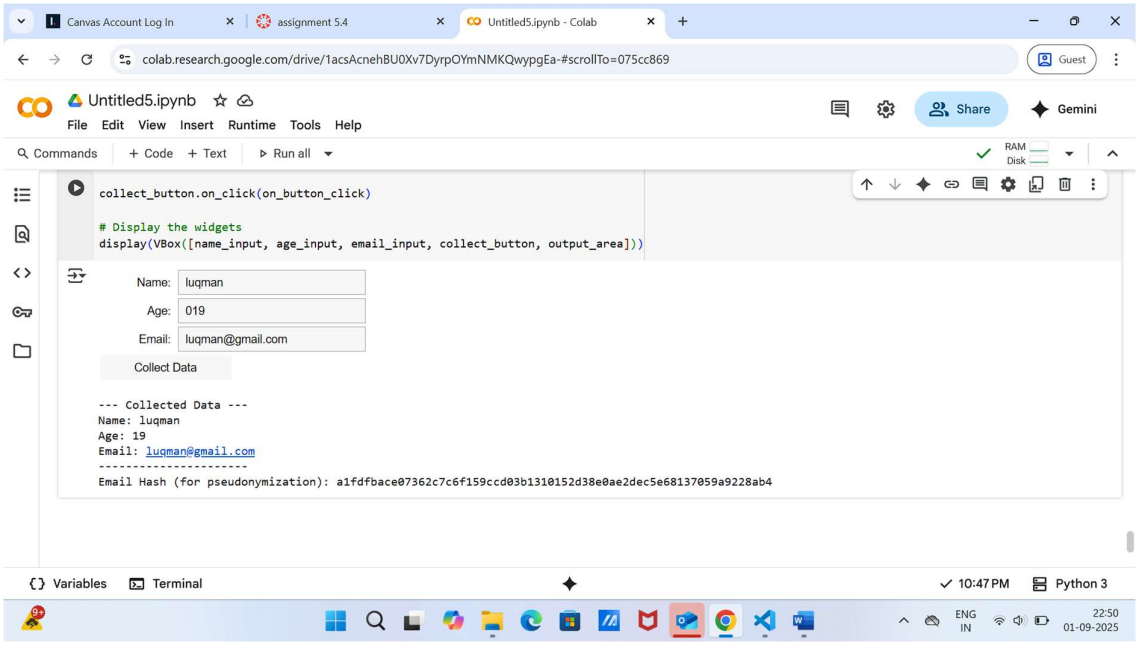
22:50
01-09-2025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collect_button.on_click(on_button_click)
# Display the widgets
display(VBox([name_input, age_input, email_input, collect_button, output_area]))
Name: luqman
Age: 019
Email: luqman@gmail.com
Collect Data
--- Collected Data ---
Name: luqman
Age: 19
Email: luqman@gmail.com
Email Hash (for pseudonymization): a1fdfbace07362c7c6f159ccd03b1310152d38e0ae2dec5e68137059a9228ab4
Variables
Terminal
10:47 PM
Python 3
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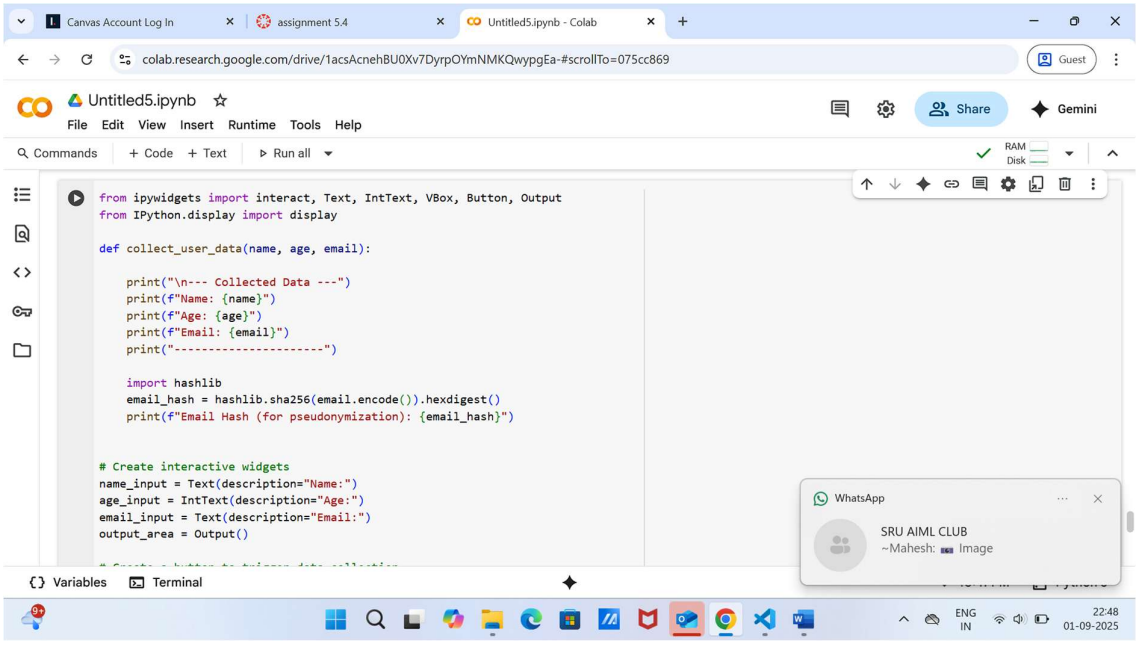
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from ipywidgets import interact, Text, IntText, VBox, Button, Output
from IPython.display import display
def collect_user_data(name, age, email):
print("\n--- Collected Data ---")
print(f"Name: {name}")
print(f"Age: {age}")
print(f"Email: {email}")
print("----------")
import hashlib
email_hash = hashlib.sha256(email.encode()).hexdigest()
print(f"Email Hash (for pseudonymization): {email_hash}")
# Create interactive widgets
name_input = Text(description="Name:")
age_input = IntText(description="Age:")
email_input = Text(description="Email:")
output_area = Output()
WhatsApp
SRU AIML CLUB
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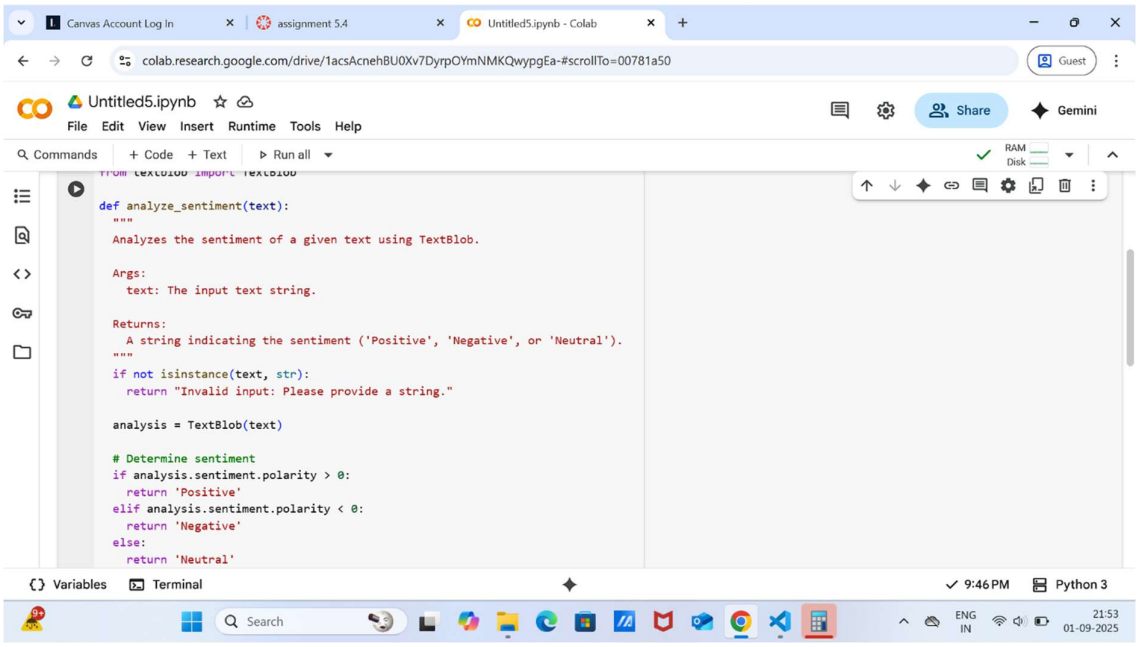
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def analyze_sentiment(text):
"""
Analyzes the sentiment of a given text using TextBlob.
Args:
text: The input text string.
Returns:
A string indicating the sentiment ('Positive', 'Negative', or 'Neutral').
"""
if not isinstance(text, str):
return "Invalid input: Please provide a string."
analysis = TextBlob(text)
# Determine sentiment
if analysis.sentiment.polarity > 0:
return 'Positive'
elif analysis.sentiment.polarity < 0:
return 'Negative'
else:
return 'Neutral'
Variables
Terminal
9:46 PM
Python 3
Search
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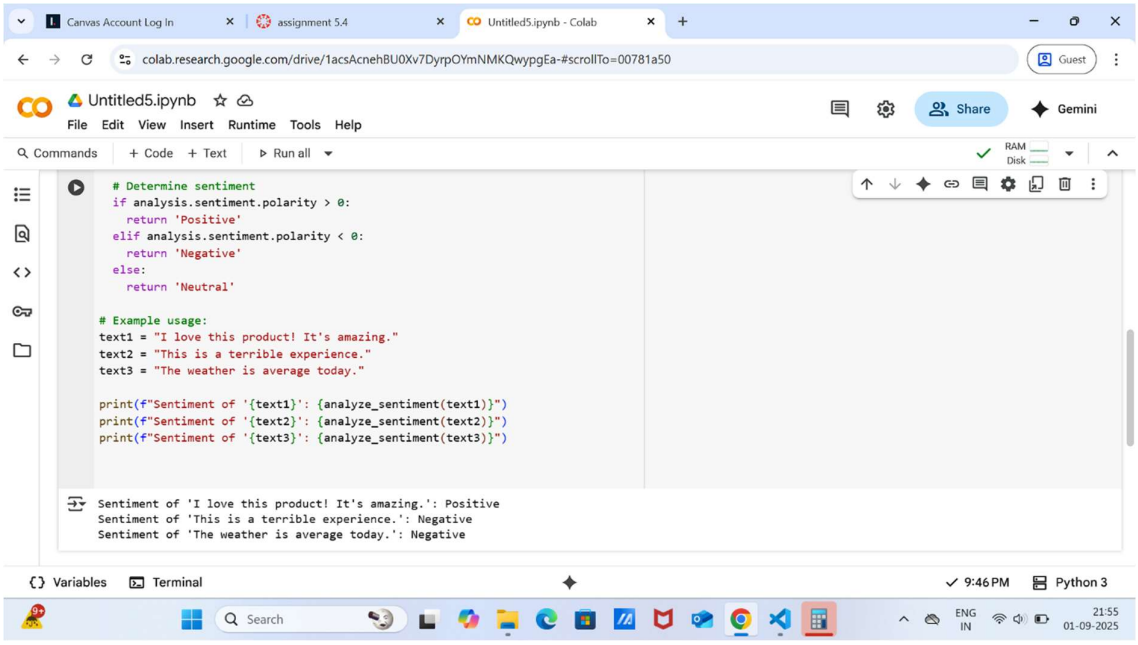
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Disk
# Determine sentiment
if analysis.sentiment.polarity > 0:
return 'Positive'
elif analysis.sentiment.polarity < 0:
return 'Negative'
else:
return 'Neutral'
# Example usage:
text1 = "I love this product! It's amazing."
text2 = "This is a terrible experience."
text3 = "The weather is average today."
print(f"Sentiment of '{text1}': {analyze_sentiment(text1)}")
print(f"Sentiment of '{text2}': {analyze_sentiment(text2)}")
print(f"Sentiment of '{text3}': {analyze_sentiment(text3)}")
Sentiment of 'I love this product! It's amazing.': Positive
Sentiment of 'This is a terrible experience.': Negative
Sentiment of 'The weather is average today.': Negative
Variables
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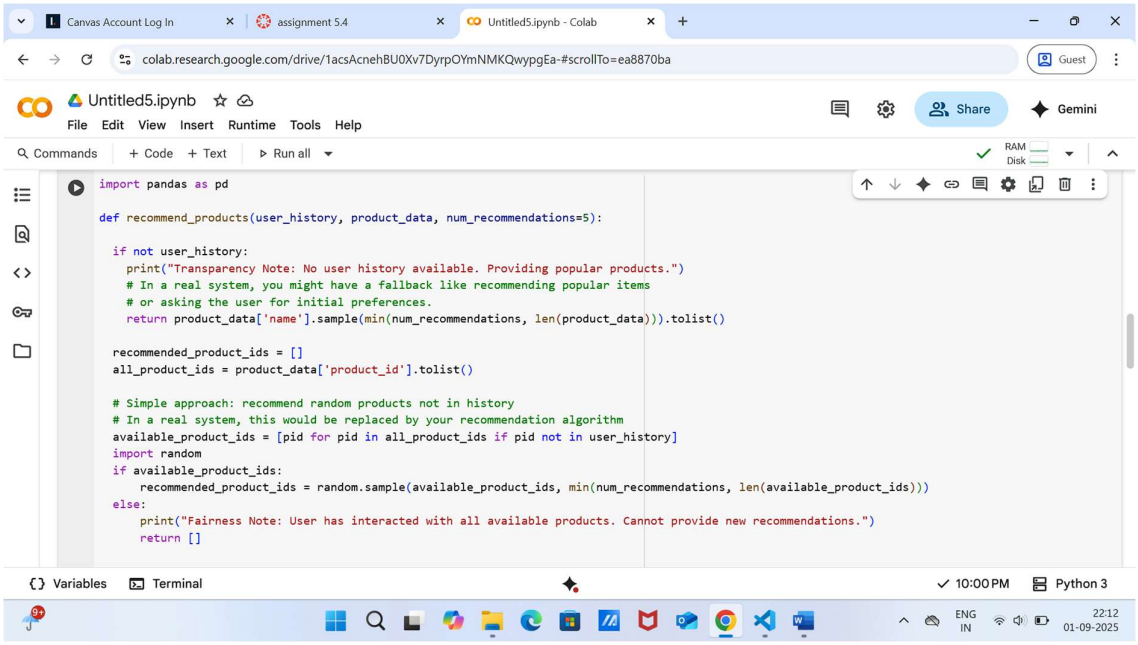
Task 3: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import pandas as pd
def recommend_products(user_history, product_data, num_recommendations=5):
if not user_history:
print("Transparency Note: No user history available. Providing popular products.")
# In a real system, you might have a fallback like recommending popular items
# or asking the user for initial preferences.
return product_data['name'].sample(min(num_recommendations, len(product_data))).tolist()
recommended_product_ids = []
all_product_ids = product_data['product_id'].tolist()
# Simple approach: recommend random products not in history
# In a real system, this would be replaced by your recommendation algorithm
available_product_ids = [pid for pid in all_product_ids if pid not in user_history]
import random
if available_product_ids:
recommended_product_ids = random.sample(available_product_ids, min(num_recommendations, len(available_product_ids)))
else:
print("Fairness Note: User has interacted with all available products. Cannot provide new recommendations.")
return []
Variables
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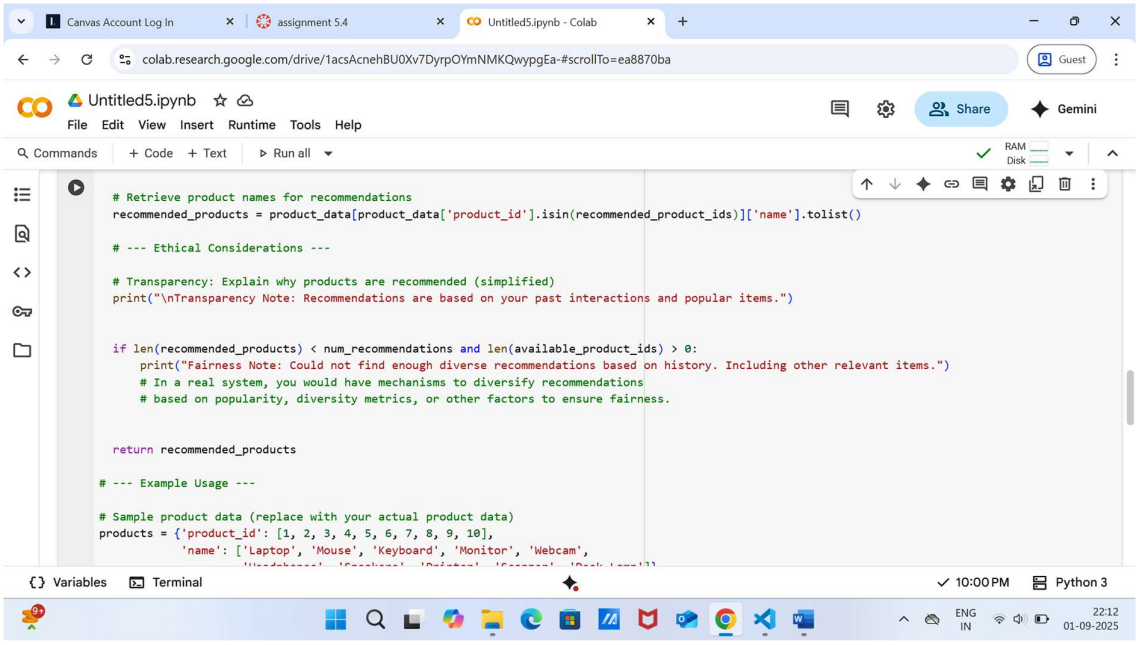
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# Retrieve product names for recommendations
recommended_products = product_data[product_data['product_id'].isin(recommended_product_ids)]['name'].tolist()
# --- Ethical Considerations ---
# Transparency: Explain why products are recommended (simplified)
print("\nTransparency Note: Recommendations are based on your past interactions and popular items.")
if len(recommended_products) < num_recommendations and len(available_product_ids) > 0:
print("Fairness Note: Could not find enough diverse recommendations based on history. Including other relevant items.")
# In a real system, you would have mechanisms to diversify recommendations
# based on popularity, diversity metrics, or other factors to ensure fairness.
return recommended_products
# --- Example Usage ---
# Sample product data (replace with your actual product data)
products = {'product_id': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
'name': ['Laptop', 'Mouse', 'Keyboard', 'Monitor', 'Webcam',
Variables
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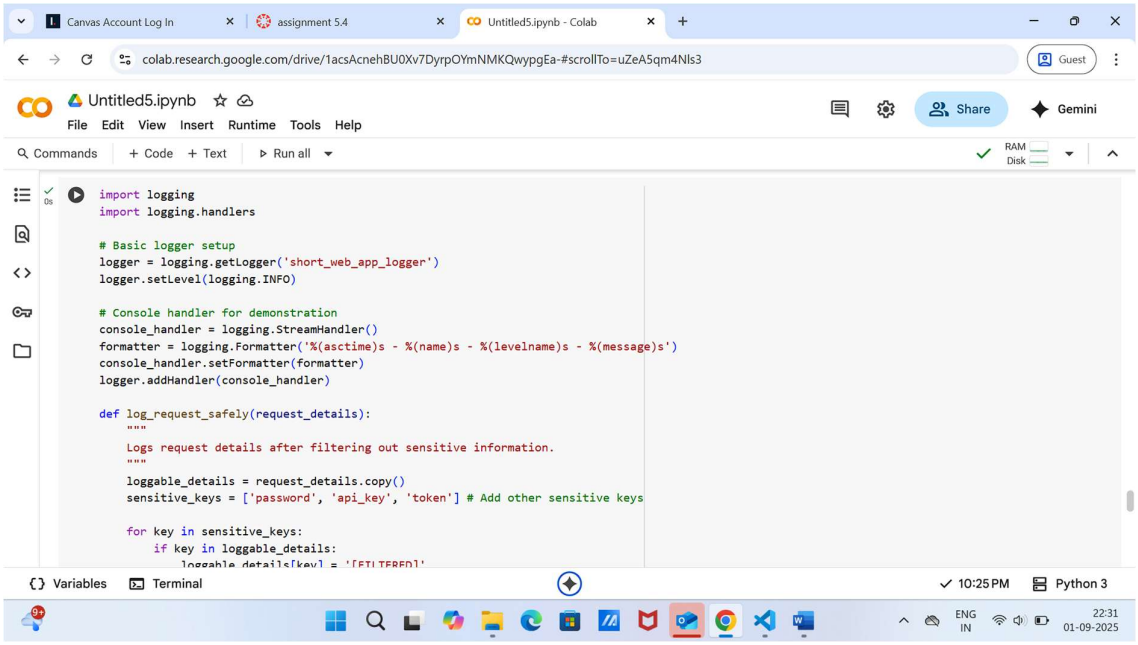
Task 4: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import logging
import logging.handlers
# Basic logger setup
logger = logging.getLogger('short_web_app_logger')
logger.setLevel(logging.INFO)
# Console handler for demonstration
console_handler = logging.StreamHandler()
formatter = logging.Formatter('%(asctime)s - %(name)s - %(levelname)s - %(message)s')
console_handler.setFormatter(formatter)
logger.addHandler(console_handler)
def log_request_safely(request_details):
"""
Logs request details after filtering out sensitive information.
"""
loggable_details = request_details.copy()
sensitive_keys = ['password', 'api_key', 'token'] # Add other sensitive keys
for key in sensitive_keys:
if key in loggable_details:
Variables
Terminal
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Python 3
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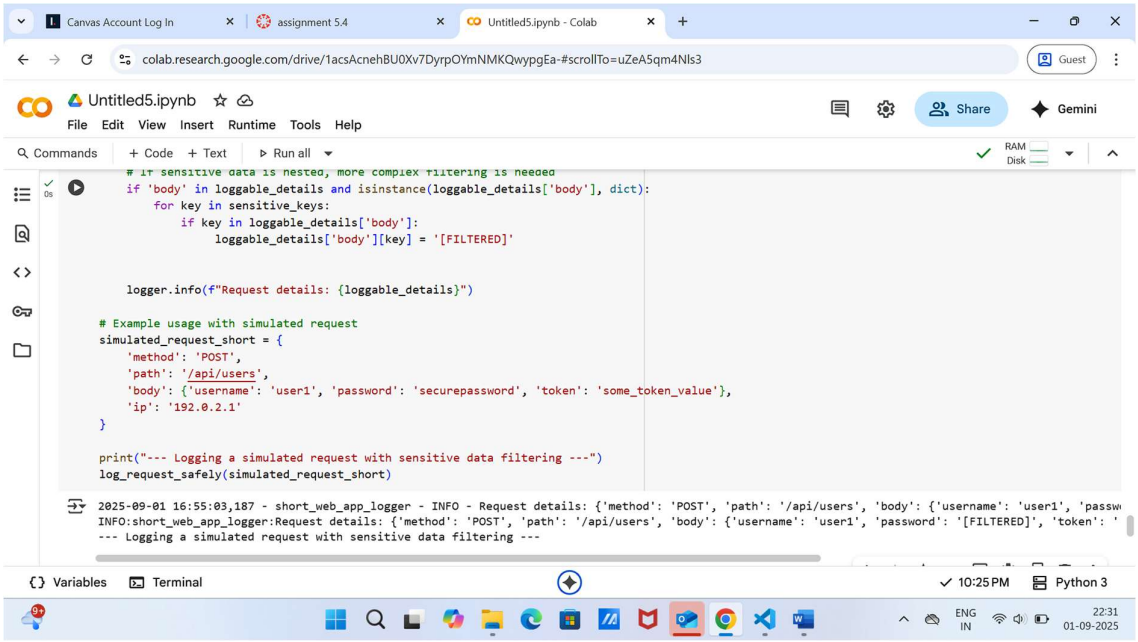
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Run all
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# If sensitive data is nested, more complex filtering is needed
if 'body' in loggable_details and isinstance(loggable_details['body'], dict):
for key in sensitive_keys:
if key in loggable_details['body']:
loggable_details['body'][key] = '[FILTERED]'
logger.info(f"Request details: {loggable_details}")
# Example usage with simulated request
simulated_request_short = {
'method': 'POST',
'path': '/api/users',
'body': {'username': 'user1', 'password': 'securepassword', 'token': 'some_token_value'},
'ip': '192.0.2.1'
}
print("--- Logging a simulated request with sensitive data filtering ---")
log_request_safely(simulated_request_short)
2025-09-01 16:55:03,187 - short_web_app_logger - INFO - Request details: {'method': 'POST', 'path': '/api/users', 'body': {'username': 'user1', 'passw
INFO:short_web_app_logger:Request details: {'method': 'POST', 'path': '/api/users', 'body': {'username': 'user1', 'password': '[FILTERED]', 'token': '
--- Logging a simulated request with sensitive data filtering ---
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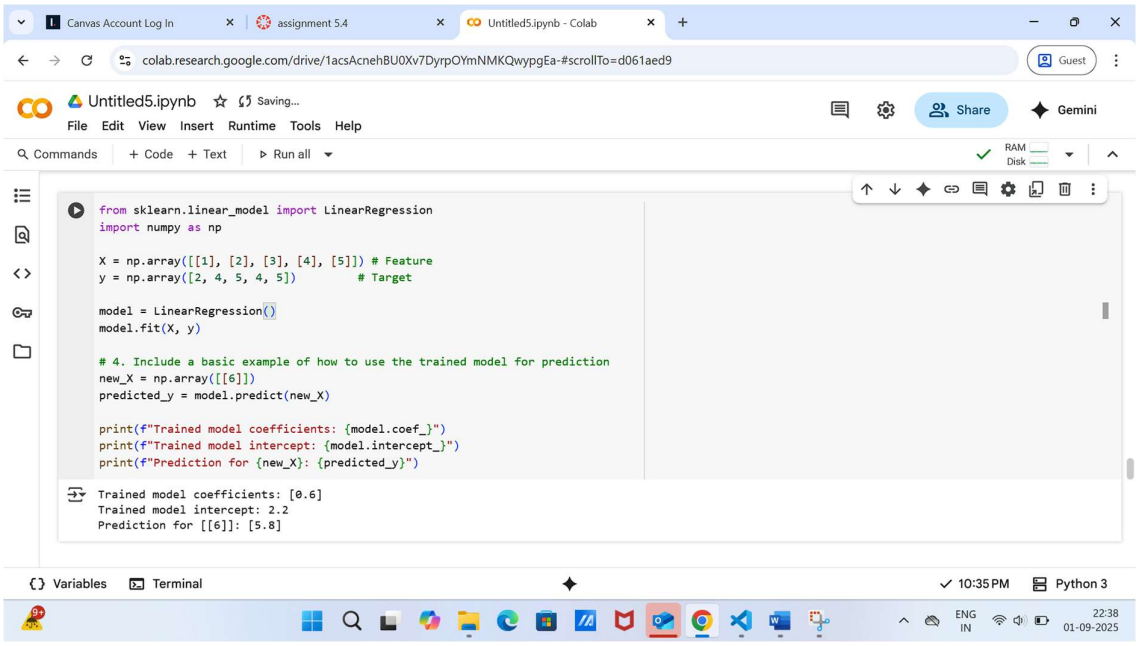
Task 5: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from sklearn.linear_model import LinearRegression
import numpy as np
X = np.array([[1], [2], [3], [4], [5]]) # Feature
y = np.array([2, 4, 5, 4, 5]) # Target
model = LinearRegression()
model.fit(X, y)
# 4. Include a basic example of how to use the trained model for prediction
new_X = np.array([[6]])
predicted_y = model.predict(new_X)
print(f"Trained model coefficients: {model.coef_}")
print(f"Trained model intercept: {model.intercept_}")
print(f"Prediction for {new_X}: {predicted_y}")
Trained model coefficients: [0.6]
Trained model intercept: 2.2
Prediction for [[6]]: [5.8]
Variables
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Python 3
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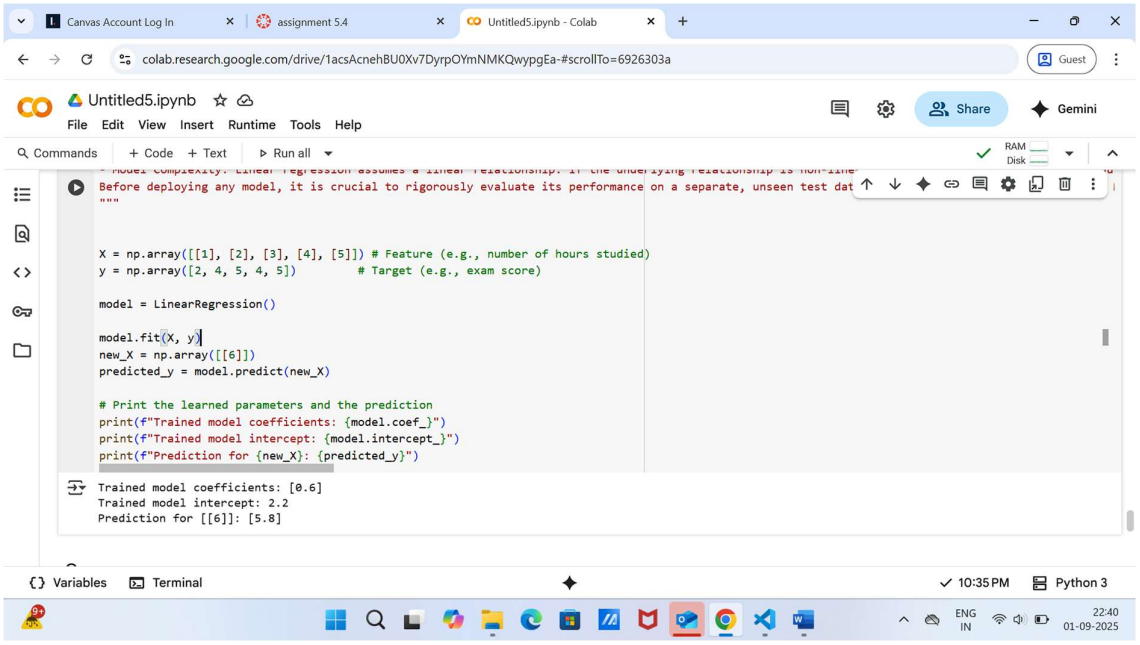
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Before deploying any model, it is crucial to rigorously evaluate its performance on a separate, unseen test dat
"""
X = np.array([[1], [2], [3], [4], [5]]) # Feature (e.g., number of hours studied)
y = np.array([2, 4, 5, 4, 5]) # Target (e.g., exam score)
model = LinearRegression()
model.fit(X, y)
new_X = np.array([[6]])
predicted_y = model.predict(new_X)
# Print the learned parameters and the prediction
print(f"Trained model coefficients: {model.coef_}")
print(f"Trained model intercept: {model.intercept_}")
print(f"Prediction for {new_X}: {predicted_y}")
Trained model coefficients: [0.6]
Trained model intercept: 2.2
Prediction for [[6]]: [5.8]
Variables Terminal
10:35 PM Python 3
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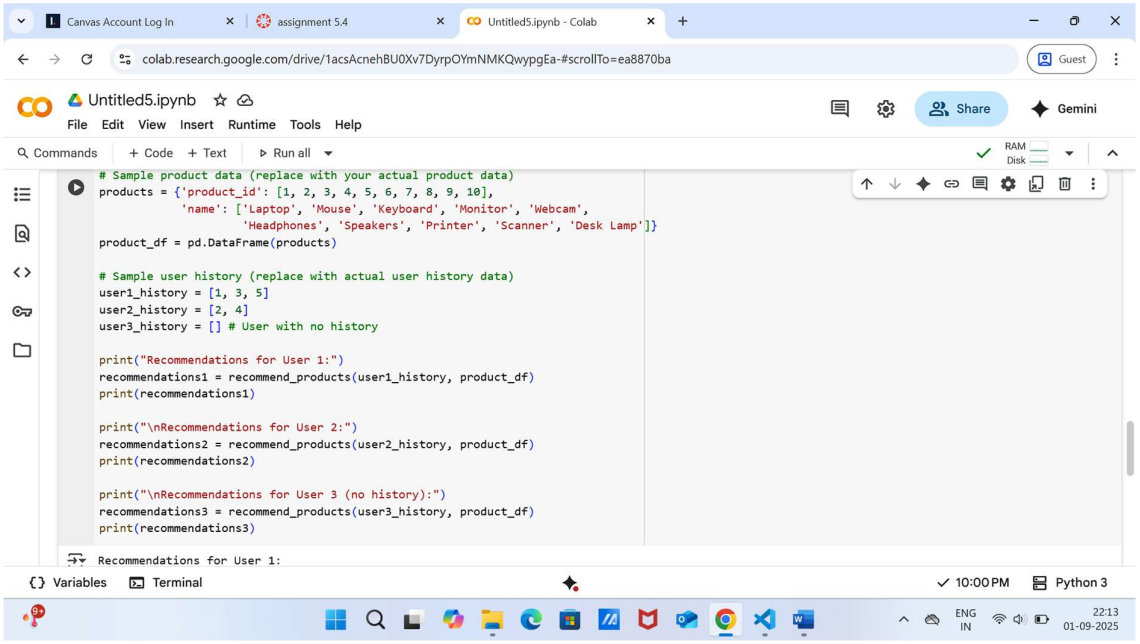
01-09-2025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Commands + Code + Text Run all
RAM Disk
# Sample product data (replace with your actual product data)
products = {'product_id': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
'name': ['Laptop', 'Mouse', 'Keyboard', 'Monitor', 'Webcam',
'Headphones', 'Speakers', 'Printer', 'Scanner', 'Desk Lamp']}
product_df = pd.DataFrame(products)
# Sample user history (replace with actual user history data)
user1_history = [1, 3, 5]
user2_history = [2, 4]
user3_history = [] # User with no history
print("Recommendations for User 1:")
recommendations1 = recommend_products(user1_history, product_df)
print(recommendations1)
print("\nRecommendations for User 2:")
recommendations2 = recommend_products(user2_history, product_df)
print(recommendations2)
print("\nRecommendations for User 3 (no history):")
recommendations3 = recommend_products(user3_history, product_df)
print(recommendations3)
Recommendations for User 1:
Variables Terminal
10:00 PM Python 3
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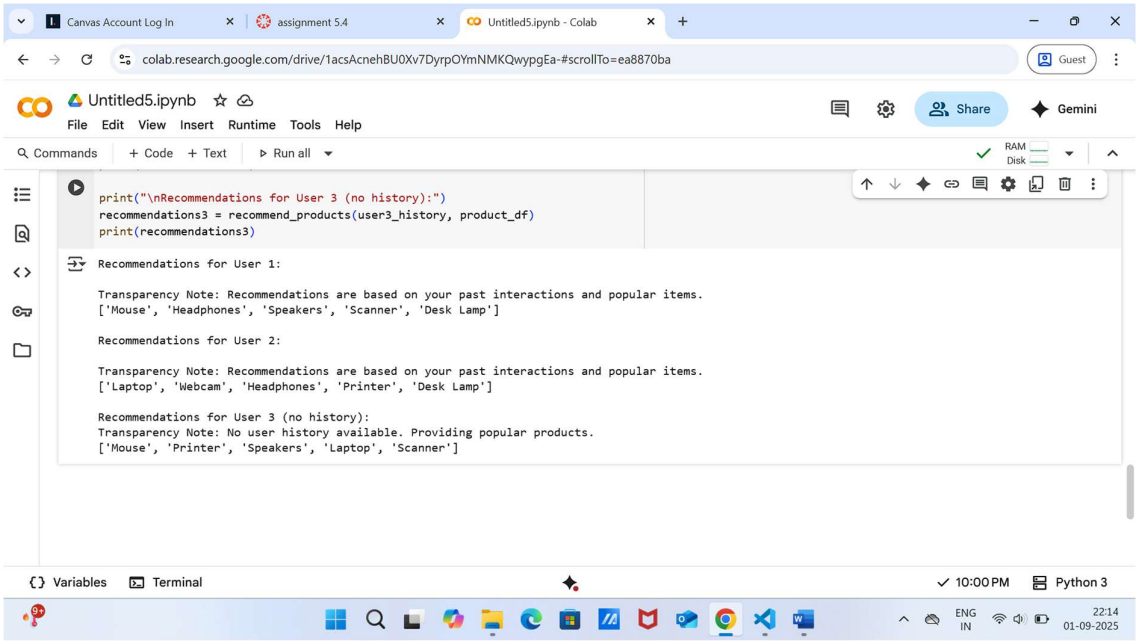
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RAM
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print("\nRecommendations for User 3 (no history):")
recommendations3 = recommend_products(user3_history, product_df)
print(recommendations3)
Recommendations for User 1:
Transparency Note: Recommendations are based on your past interactions and popular items.
['Mouse', 'Headphones', 'Speakers', 'Scanner', 'Desk Lamp']
Recommendations for User 2:
Transparency Note: Recommendations are based on your past interactions and popular items.
['Laptop', 'Webcam', 'Headphones', 'Printer', 'Desk Lamp']
Recommendations for User 3 (no history):
Transparency Note: No user history available. Providing popular products.
['Mouse', 'Printer', 'Speakers', 'Laptop', 'Scanner']
Variables
Terminal
10:00 PM
Python 3
ENG
IN
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Task 4: